*updated: 7/25*

# Curriculum Vitae: Chad McKell

## ABOUT


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| Position | Research Engineer |
| Company | Sigma Connectivity, on assignment at Meta |
| Address | 10301 Willows Road NE, Redmond, WA 98052 |
| Phone | +1 650 308-7837 |
| Email | `cmckell@meta.com` |
| Website | `chadmckell.com` |


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| Research | My research spans mathematical modeling and numerical simulation of physical systems with an emphasis on acoustics and audio. Recent projects include geometric boundary modeling for acoustic wave simulation; non-spherical harmonics for sound capture; and elastic wave modeling for sound synthesis and auditory biophysical analysis. Past projects include underwater acoustic sensing and optical standing-wave trapping. |

## EDUCATION

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| 9/19– | **University of California San Diego**, Ph.D. in Computer Music<br>Specialized in Computational Acoustics and Applied Mathematics<br>Dissertation: *Conformal Symmetry for Exterior Wave Problems*<br>Advisors: Albert Chern (Computer Science), Miller Puckette (Music/IRCAM) |
| 9/16–10/17 | **University of Edinburgh**, M.S. in Acoustics and Music Technology |
| 8/09–12/15 | **Wake Forest University**, M.S. in Physics |
| 6/02–8/09 | **Brigham Young University**, B.S. in Biophysics |

## EMPLOYMENT

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|---|---|
| 9/24– | **Sigma Connectivity, on assignment at Meta**, Research Engineer |
| 3/24–6/24 | **University of California San Diego**, Teaching Assistant in Computer Science |
| 6/23–2/24 | **Meta, Reality Labs Research**, Research Scientist Intern |
| 8/21–3/22 | **Meta, Reality Labs Research**, Research Intern |
| 9/19–6/23 | **University of California San Diego**, Teaching Assistant/Researcher in Music |
| 7/18–7/19 | **Applied Research in Acoustics**, R&D Scientist |
| 5/18–5/18 | **Moog Music (self-employed)**, Audio Software Developer |
| 4/17–9/17 | **Lofelt, acquired by Meta in 2022 (self-employed)**, Acoustics Researcher |
| 10/14–8/16 | **J.P. Morgan/Neovest (via ConsultNet)**, Software Development Engineer in Test |
| 8/12–12/12 | **University of North Carolina School of the Arts**, Adjunct Instructor of Physics |
| 9/09–9/11 | **Wake Forest University**, Teaching Assistant in Physics |
| 9/08–6/09 | **Brigham Young University**, Tutorial Lab Assistant in Physics |
| 8/07–3/09 | **Brigham Young University**, Research Assistant in Philosophy |

## RESEARCH ACTIVITIES

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|---|---|
| 9/24– | **Sigma Connectivity, on assignment at Meta**, Research Engineer<br>Redmond, Washington. Affiliation: Reality Labs Research Audio. Research topics: *spatial audio, software engineering*. Research summary: Build novel spatial audio technology for Reality Labs Research at Meta. Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra. I work onsite at Meta as a vendor employed by Sigma Connectivity. |

## RESEARCH ACTIVITIES CONT.

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|---|---|
| 9/19–9/24 | **University of California San Diego**, Ph.D. Student<br>La Jolla, California. Affiliations: Department of Music, Center for Visual Computing. Research topics: *computational acoustics, applied mathematics.* Research summary: developed mathematical and computational methods for solving exterior wave problems, with applications in virtual acoustics. Committee: Albert Chern (co-chair, Computer Science), Miller Puckette (co-chair, Music/IRCAM), Melvin Leok (Mathematics), Shahrokh Yadegari (Music), Stefan Bilbao (IRCAM), Sebastian Prepeliță (Meta). |
| 6/23–2/24 | **Meta, Reality Labs Research**, Research Scientist Intern<br>Redmond, Washington. Affiliation: Reality Labs Research Audio. Research topics: *spatial audio, software engineering.* Research summary: Built novel spatial audio technology for applications in virtual and augmented reality. Supervisor: Sebastian Prepeliță. Team Lead: Ravish Mehra. |
| 8/21–3/22 | **Meta, Reality Labs Research**, Research Intern<br>La Jolla, California. Research description: see above. |
| 7/18–7/19 | **Applied Research in Acoustics**, R&D Scientist<br>Culpeper, Virginia. Research topics: *underwater acoustic sensing, acoustic beamforming.* Research summary: developed physics-based signal processing algorithms for object detection and classification in naval sonar systems. Team Lead: Jonathan Botts. |
| 1/17–8/17 | **University of Edinburgh**, Master's Student<br>Edinburgh, Scotland. Affiliations: Reid School of Music, Edinburgh College of Art. Research topics: *computational acoustics, elastodynamics, numerical simulation.* Research summary: developed numerical simulations of structural acoustic vibrations for commercial haptic devices. My thesis was partially funded by Lofelt, a Berlin-based haptic feedback company acquired by Meta in 2022. Advisor: Stefan Bilbao. |
| 1/10–9/13 | **Wake Forest University**, Master's Student<br>Winston-Salem, North Carolina. Affiliation: Department of Physics. Research topics: *optical trapping, laser beam characterization, fluid dynamics.* Research summary: simulated, fabricated, and analyzed standing-wave optical traps for Brownian particle tracking. Advisor: Keith Bonin. |
| 8/07–8/09 | **Brigham Young University**, Undergraduate Student<br>Provo, Utah. Affiliation: Department of Physiology and Developmental Biology. Research topics: *biophysics, scanning probe microscopy.* Research summary: investigated structural effects of isoflurane on artificial lipid bilayers using atomic force microscopy. Advisor: David Busath. |

## TEACHING EXPERIENCE

### ***As Instructor***

#### **UNCSA**

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| SCI 1100 | General Physics. Fall 2012 (1 term). |

## TEACHING EXPERIENCE CONT.

### ***As TA***

**UCSD**

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| CSE 291 | Physics Simulation. Spring 2024 (1 term). |
| MUS 5 | Sound in Time. Spring 2020 (1 term). |
| MUS 6 | Electronic Music. Fall 2020 (1 term). |
| MUS 15 | Popular Music: David Bowie. Winter 2021 (1 term). |
| MUS 15 | Popular Music: Video Game Music. Winter 2020 (1 term). |
| MUS 171 | Computer Music I. Winter 2022 (1 term). |
| MUS 172 | Computer Music II. 2021–2022 (2 terms). |

**WFU**

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|---|---|
| PHY 113 | General Physics I (Mechanics). 2009–2011 (4 terms). |
| PHY 114 | General Physics II (E&M). Fall 2010 (1 term). |

### ***As Tutor***

**BYU**

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| PHSCS 105 | General Physics 1 (Mechanics). 2008–2009 (2 terms). |
| PHSCS 106 | General Physics 2 (E&M). Winter 2009 (1 term). |
| PHSCS 121 | Principles of Physics 1 (Mechanics). 2008–2009 (2 terms). |
| PHSCS 123 | Principles of Physics 2 (Waves/Thermo). W/Sp 2009 (2 terms). |
| PHSCS 220 | Principles of Physics 3 (E&M). W/Sp 2009 (2 terms) |

## PUBLICATIONS

### Manuscripts in Progress

(1) **C. McKell**, M. Nabizadeh, S. Wang, and A. Chern, "Wave simulations in infinite spacetime".

*Simulating wave propagation on an infinite domain has been a long-standing computational challenge. Conventional approaches to this problem only produce wave simulations on a small subset of the infinite domain. Using the fact that wave propagation on an infinite Minkowski spacetime is equivalent to wave propagation on a bounded Minkowski spacetime under a Kelvin-like transformation, we simulate wave propagation on the entire infinite domain using a finite discretization of the bounded domain with no additional loss of accuracy from the transformation.*

(2) **C. McKell**, *Conformal Symmetry for Exterior Wave Problems*, Ph.D. Dissertation, University of California San Diego. Defense planned for Fall 2025. Advisors: Albert Chern and Miller Puckette.

*This work presents novel geometric methods for handling open exterior boundaries in wave simulations. First, I discuss extensions to the reflectionless discrete perfectly matched layer for the wave and Helmholtz equations. Then, I demonstrate that the conformal invariance of the wave equation under a Kelvin transform in Minkowski spacetime allows one to convert an infinite domain problem into a bounded domain problem that can be solved using standard numerical methods with no additional loss of accuracy introduced by the transform. I conclude by discussing parallel computing strategies for the geometric boundary models, applications in architectural acoustics and binaural audio, and future work in obstacle boundary flattening.*

## PUBLICATIONS CONT.

### Journal Articles

(3) **C. McKell** and K. Bonin, "Optical corral using a standing-wave Bessel beam," *Journal of the Optical Society of America B*, Vol. 35, No. 8, 1910–1920, 2018.

### Conference Proceedings

(4) **C. McKell**, "Sonification of optically-ordered Brownian motion," In Proceedings of the International Computer Music Conference (ICMC), Utrecht, Netherlands, September 2016.

### Master's Theses

(5) **C. McKell**, *Real-Time Physical Modeling for Haptic Feedback Rendering*, Master's Thesis, University of Edinburgh, Acoustics and Audio Group, 2017. Advisor: Stefan Bilbao.

(6) **C. McKell**, *Confinement and Tracking of Brownian Particles in a Bessel Beam Standing Wave*, Master's Thesis, Wake Forest University, Department of Physics, 2015. Advisor: Keith Bonin.

### Technical Reports

(7) **C. McKell**, H. Conley, and D. Busath, "AFM study of structural changes in supported planar DPPC bilayers containing general anesthetic isoflurane," Brigham Young University, Paper 827, 2010.